Build an end-to-end machine learning solution that predicts customer churn and converts model outputs into actionable retention insights. The intern must demonstrate data auditing, EDA, feature engineering, leakage prevention, model development, class-imbalance handling, hyperparameter tuning, robust evaluation, explainability, and business interpretation.

A subscription service provider wants to identify customers at high risk of leaving during the next retention cycle. The company can contact only a limited percentage of customers, so the project must focus on reliable ranking of risk and explainable predictions rather than accuracy alone.

A. Data Audit & Preparation

- Inspect data types, missing values, duplicates, invalid values, outliers, and target distribution.
- Document cleaning decisions; do not silently remove observations.
- Build a reproducible preprocessing pipeline for numeric and categorical variables.
- Treat customer_id as an identifier, not a predictive feature.

B. Exploratory Data Analysis

- Analyze churn by contract, tenure, charges, satisfaction, complaints, late payments, usage, and payment method.
- Investigate nonlinear relationships and meaningful interactions.
- Identify unusual observations and discuss whether they are errors or legitimate cases.
- Create at least 6 informative visualizations with concise interpretations.

C. Feature Engineering

- Create at least 5 justified features, such as charge-per-tenure, payment-risk indicators, complaint intensity, or interaction terms.
- Explain the business/data rationale for each engineered feature.
- Demonstrate that engineered variables do not leak the target.

D. Modeling

- Build at least three classification models, including Logistic Regression and two stronger alternatives such as Random Forest, Gradient Boosting, XGBoost/LightGBM, or HistGradientBoostingClassifier.
- Use a stratified train/test split and keep the final test set untouched until final evaluation.
- Use cross-validation and hyperparameter tuning.
- Compare baseline and tuned models.

E. Class Imbalance & Evaluation

- Measure target imbalance and justify the selected treatment.
- Consider class weights, threshold tuning, or resampling; any resampling must occur inside the training/CV pipeline.
- Report precision, recall, F1, ROC-AUC, PR-AUC, and a confusion matrix.
- Explain why accuracy alone is insufficient.
- Select an operating threshold using a stated business assumption rather than automatically using 0.50.

F. Explainability

- Use SHAP, permutation importance, or another defensible method.

- Provide global feature importance and at least three individual customer-level explanations.
- Separate predictive association from causation.

G. Retention Intervention Simulation

- Rank customers by predicted churn probability and assume the company can contact only the top 20%.
- Calculate how many actual churners are captured by this top-risk group.
- Compare this with randomly contacting 20% of customers.
- Run a simple cost-benefit simulation using clearly stated hypothetical contact cost and retained-customer value assumptions.

H. Final Analysis

- Explain the model-selection rationale using documented metrics.
- Identify major churn-risk patterns supported by the data.
- Propose 3–5 retention actions linked to observed patterns, clearly separating evidence from business assumptions.

Advanced Requirements

- Use Pipeline and ColumnTransformer where appropriate.
- Use cross-validation for model selection and report the search strategy.
- Show precision/recall behavior across multiple classification thresholds.
- Include a calibration check or calibration curve and discuss probability reliability.
- Perform at least one robustness check, such as fold-to-fold variation or an outlier sensitivity test.
- Document random seeds, package versions, assumptions, and execution steps.

Deliverables

- 01_data_science_notebook.ipynb — complete executable analysis.
- 02_cleaned_dataset.csv — cleaned/processed data where applicable.
- 03_model_artifact — saved final model/pipeline.
- 04_business_report.pdf — 5–8 page professional report.
- 05_README.md — setup, assumptions, execution instructions, and output descriptions.

Suggested Deadline & Milestones

Milestone	Target	Expected Output
Data audit + EDA	Day 2	Data-quality report + visualizations
Feature engineering + baseline	Day 4	Feature set + baseline model
Tuning + imbalance handling	Day 6	Cross-validated comparison
Explainability + intervention simulation	Day 8	Explanations + top-20% analysis
Final submission	Day 10	Notebook, model, report, README